Is Prosody Lost in Translation?

Fine-Grained Cross-Lingual Prosody Similarity Across Languages

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INTRODUCTION

Speech translation mainly preserves words. We study how much of the source prosody remains in professionally dubbed speech.

For example, "oh my god" and "dios mío" can express surprise or frustration depending on pitch and energy.

We compare aligned word-level contours across German–English, English–Spanish and English–French.

RESEARCH QUESTIONS

How similar are pitch and energy after translation?
Which word classes contribute most?
How closely does utterance timing correspond?

DATA

Professionally dubbed TV dialogue, mined with SpeechVecAlign [1]. We filter each language pair separately.

Stage	DE–EN	EN–ES	EN–FR
Raw pairs	100,000	100,000	100,000
Audio quality	41,610	42,974	42,045
One speaker	28,041	28,990	29,448
Same meaning	20,964	22,598	20,342
Valid pitch	**16,335**	**16,939**	**15,957**
Hours[1]	36.7	37.9	35.7

Overall: 49,231 pairs and 110.3 hours[1]

[1] Source + target hours. Final counts follow all filters (Tables 2–3 in the paper).

METHOD

We transcribe both sides with word timestamps [5] and obtain bilingual word links with FastAlign [6].

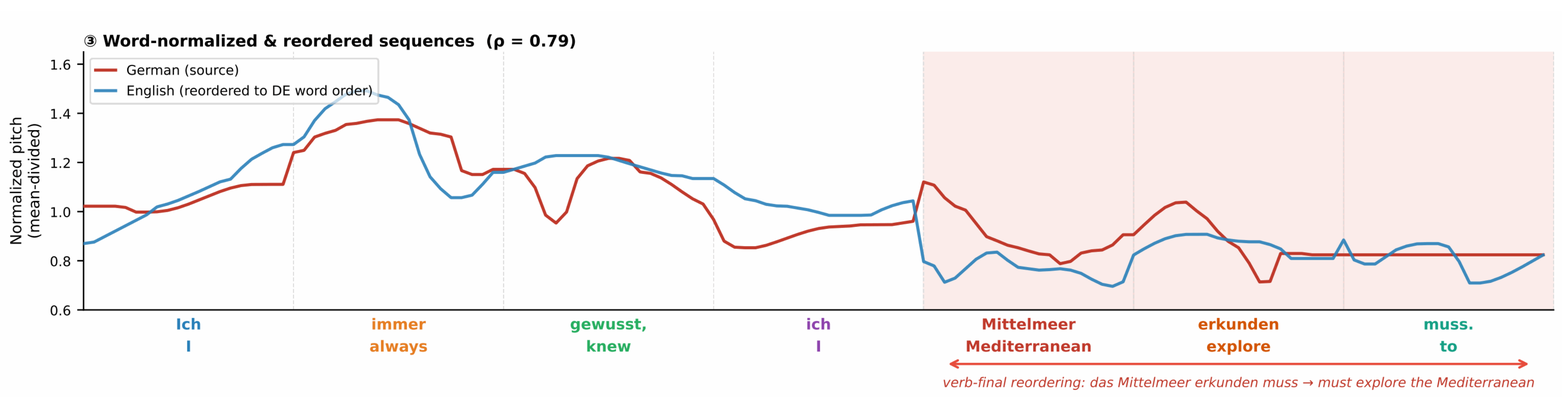
We extract pitch with pYAAPT [7] and energy with librosa [8]. We resample each aligned cluster to 20 frames, reorder target clusters and mean-normalize the contours.

We compute Spearman ρ per utterance and compare it with a shuffled-word baseline.

GERMAN–ENGLISH EXAMPLE FROM FIGURE 2

DE: "Ich habe immer gewusst, dass ich weiter das Mittelmeer erkunden muss."

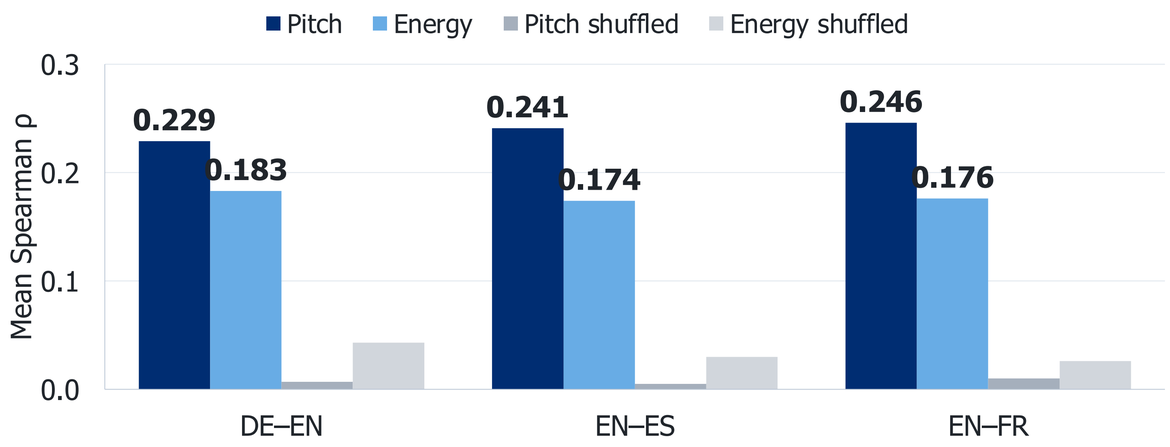
EN: "I always knew I had to return to explore the Mediterranean Sea."



The English segments follow the German word order before comparison.
This example has ρ = 0.79. Corpus means are much lower (below).

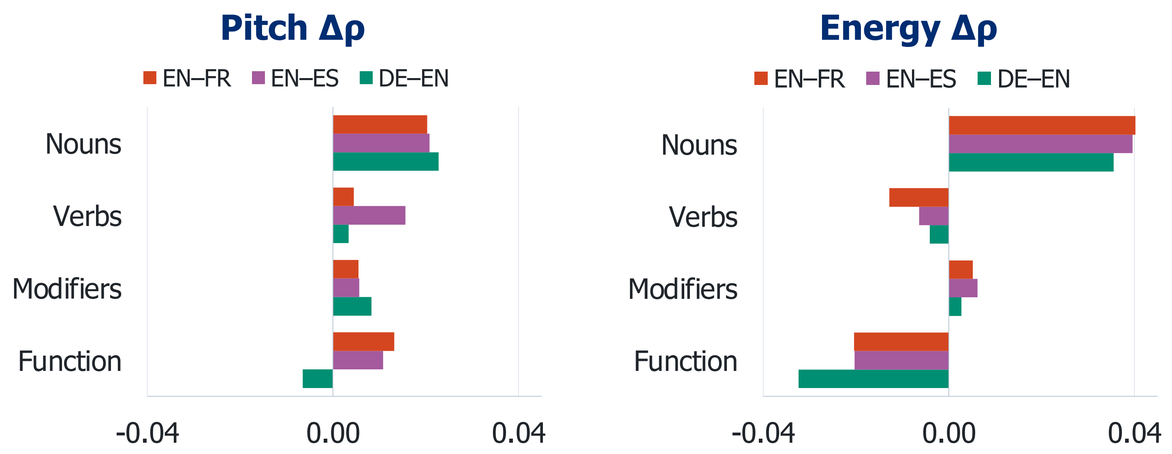
RESULTS

RQ1 Pitch and energy show positive correspondence



Pitch ρ = 0.229–0.246; energy ρ = 0.174–0.183.
Both exceed shuffled baselines ($p < 10^{-80}$).

RQ2 Nouns have the largest ablation effect



Δρ = ρ(all words) − ρ(with class removed).
Removing nouns lowers energy ρ by 0.036–0.040.

RQ3 Timing is strongly correlated in dubbed speech

Dubbing requires temporal synchronization.

Duration: 0.884–0.891
Phoneme count: 0.871–0.885

CONCLUSIONS

Prosodic structure is partly shared across these language pairs. Pitch corresponds more strongly than energy.
Word alignment helps us compare local prosodic patterns across different word orders.

LIMITATIONS / CONTACT

Dubbing is a proxy for prosody transfer. We study three European pairs and a limited set of features. The corpus has licensing restrictions.

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jackxie.com/research/cross-lingual-prosody/

[1] Meng & Koehn, 2025 [2] Kumar et al., 2023 [3] Bredin et al., 2020 [4] Duquenne et al., 2023
[5] Louradour, 2023 [6] Dyer et al., 2013 [7] Kasi & Zahorian, 2002 [8] McFee et al., 2015